

Notes: Hsieh and Klenow (QJE, 2014)
The Life Cycle of Plants in India and Mexico

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1 Big picture

- **Research question.** Why do manufacturing plants in India and Mexico remain small as they age, while U.S. plants grow dramatically over the life cycle? Can this difference in plant dynamics help explain lower aggregate manufacturing productivity in developing countries?
- **Core empirical fact.** In the United States, old surviving plants are much larger than young plants. In India and Mexico, old plants grow far less. The typical 40-year-old U.S. plant employs more than seven times as many workers as a plant younger than five years, while old plants in India and Mexico are only around twice as large or less.
- **Economic interpretation.** Plants grow with age partly by accumulating organization capital: process know-how, product quality, managerial systems, customer relationships, supplier networks, and access to domestic and foreign markets. Slow life-cycle growth suggests lower post-entry investment in these intangibles.
- **Central mechanism.** Barriers that are especially costly for large or high-productivity plants reduce the payoff to investing in productivity. Examples include size-dependent enforcement of taxes and regulations, managerial frictions, financing constraints, land constraints, transport costs, or other distortions that make expansion unattractive.
- **Productivity implication.** If plants do not improve much after entry, the economy has fewer large, productive incumbents. Entrants face less competition, average plant size is smaller, and aggregate manufacturing TFP can be substantially lower.
- **Quantitative headline.** Simple general-equilibrium calculations suggest that moving from the U.S. life cycle to the Indian or Mexican life cycle can plausibly lower aggregate manufacturing productivity by about 25 percent.
- **Important distinction.** The paper is not mainly about entry rates or exit rates. India and Mexico do not simply lose old plants faster. Rather, surviving plants do not expand nearly as much with age.
- **Takeaway.** Development is not only about reallocating resources among existing plants. It is also about whether productive establishments have the incentive and ability to grow over their life cycle.

2 Incremental contribution of the paper

- **From static misallocation to dynamic productivity.** Hsieh and Klenow (2009) showed that static misallocation across plants can account for a sizable portion of TFP gaps. This paper asks why average plant productivity itself may be lower in poor countries.
- **Life-cycle evidence using broad microdata.** The analysis uses establishment-level manufacturing data that include both formal and informal producers. This is crucial because new and small plants are often missing from standard manufacturing surveys.
- **Cross-country comparison of plant aging.** The paper compares plant employment, exit, cohort size, productivity, and revenue productivity over age in the United States, India, and Mexico.

- **Separating survival from growth.** The paper shows that the U.S.-India/Mexico contrast is not mainly due to higher exit in developing countries or a pure selection effect among survivors.
- **Bringing revenue productivity into life-cycle analysis.** The paper uses average revenue products of labor and capital to infer whether more productive plants face higher distortions. The steep relation between productivity and revenue productivity in India and Mexico is key evidence.
- **General-equilibrium accounting.** The paper translates life-cycle productivity differences into aggregate TFP effects using simple Melitz/Atkeson–Burstein style models.
- **A dynamic misallocation channel.** The paper argues that distortions against larger or more productive plants can depress not only the static allocation of inputs today, but also the dynamic incentive to build plant-specific productivity.
- **Policy relevance.** The findings suggest that barriers to growth after entry can be as important as barriers to entry. A country can have many firms but still low productivity if firms do not scale and improve.

3 Data and measurement

3.1 United States

The U.S. data come from the Manufacturing Census, which is conducted every five years. The paper uses censuses from 1992, 1997, and 2002 for the main life-cycle analysis because these years have the most complete imputed age information.

- The census covers manufacturing establishments with paid employees.
- Age is imputed from the first year in which an establishment appears in the census.
- The variables include employment, wage bill, value added, book capital, industry, and establishment identifiers.
- Administrative records are kept because the paper’s main outcome is employment rather than detailed productivity dispersion.

Interpretation. The U.S. data provide the benchmark life cycle: entrants are small, but survivors expand strongly as they age.

3.2 India

The Indian data combine the Annual Survey of Industries (ASI) and the National Sample Survey of unorganized manufacturing (NSS).

- The ASI covers formally registered manufacturing establishments and includes large formal plants with certainty.
- The NSS covers small, informal, and unorganized establishments.

- The main combined ASI-NSS years are 1989–1990, 1994–1995, and 2010–2011, because these are years with age information for the informal sector.
- Establishment age is based on the self-reported year of initial production.
- Employment includes paid workers, contract workers, unpaid workers, and full-time-equivalent workers where needed.

Interpretation. Combining ASI and NSS is essential because most Indian manufacturing establishments, and a large share of employment, are informal.

3.3 Mexico

The Mexican data come from the Economic Census in 1998, 2003, and 2008.

- The census covers fixed establishments, but excludes street vendors.
- Establishment age is self-reported as the year the establishment began operation, regardless of ownership changes.
- Variables include paid and unpaid workers, hours worked, wage bill, nonwage compensation, capital, value added, and industry.
- The data do not have establishment identifiers that allow the authors to link individual plants across censuses.

Interpretation. Mexico is an intermediate case: plants grow more than in India, but their growth flattens relative to the U.S. benchmark.

3.4 Main empirical objects

The paper focuses on several empirical objects:

- **Employment by age:** average employment per operating plant in each age bin.
- **Exit by age:** whether old plants are rare because they exit faster.
- **Employment shares by age:** how much aggregate employment is accounted for by plants of different ages.
- **Cohort growth:** how average employment changes for cohorts as they age.
- **Productivity by age:** inferred plant productivity relative to aggregate TFP.
- **Revenue productivity by age and productivity:** whether more productive plants face higher average revenue products, interpreted as distortions or wedges.

4 Methodology and empirical specifications

4.1 Cross-sectional life-cycle profiles

The simplest profile compares average plant employment across age bins within a census year. Employment in the youngest group is normalized to one. This gives the striking Figure I fact: old U.S. plants are much larger than young U.S. plants, but old plants in India and Mexico are not.

The cross-section is informative but not definitive, because older cohorts may have been born in different macroeconomic or industrial environments. The paper therefore supplements it with cohort-based and steady-state calculations.

4.2 Exit and cohort-size adjustment

The paper checks whether small old plants in India and Mexico could simply reflect higher exit of growing plants or fewer surviving old plants. It computes age-specific exit rates and employment shares by age.

The key result is that exit rates in India and Mexico are not generally higher than in the United States. Old plants also account for a much smaller share of employment in India and Mexico than in the United States. Thus, the main issue is not merely that old plants disappear; it is that old plants do not become large.

4.3 Following cohorts over time

The paper estimates life-cycle growth by comparing cohorts in repeated cross-sections. For example, it compares the average employment of a cohort at one age in one census to the average employment of the same cohort at a later age in a later census.

This approach shows that:

- the U.S. life cycle remains steep when measured over time;
- Mexico remains intermediate;
- India's post-1994 life-cycle growth is modest but positive, while the pre-1994 pattern is much weaker.

4.4 Productivity and revenue-productivity inference

To interpret employment growth, the paper uses a monopolistic competition framework. Aggregate manufacturing output is a CES aggregate of plant-level varieties:

$$Y = \left[\sum_a \sum_{i=1}^{N_a} Y_{a,i}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}},$$

where a indexes age, i indexes plants, and $\sigma > 1$ is the elasticity of substitution.

Plant output is Cobb–Douglas:

$$Y_{a,i} = A_{a,i} K_{a,i}^\alpha L_{a,i}^{1-\alpha}.$$

Plants may face output, capital, and labor wedges. A simplified profit expression is

$$\pi_{a,i} = (1 - \tau_{Y,a,i}) P_{a,i} Y_{a,i} - (1 + \tau_{L,a,i}) w L_{a,i} - (1 + \tau_{K,a,i}) R K_{a,i}.$$

The average product wedge $\tau_{a,i}$ is proportional to a geometric average of the revenue products of capital and labor:

$$\frac{\tau_{a,i}}{\bar{\tau}} \propto \left(\frac{P_{a,i} Y_{a,i}}{K_{a,i}} \right)^\alpha \left(\frac{P_{a,i} Y_{a,i}}{L_{a,i}} \right)^{1-\alpha}.$$

Interpretation. A high revenue product means that the plant is small relative to its productivity. If more productive plants have high revenue products, then resources are not flowing sufficiently to the plants where they would be most productive.

4.5 General-equilibrium simulations

The paper then asks how much aggregate TFP changes when the economy has the Indian or Mexican life-cycle productivity profile instead of the U.S. profile. The baseline models vary:

- whether entry is fixed or free;
- whether entrant quality is fixed or endogenous;
- whether average product variation reflects overhead costs, adjustment costs, or revenue-tax-like distortions.

Aggregate TFP is decomposed into three broad components:

$$\text{TFP} = \frac{Y}{L} = \bar{A} \cdot N^{1/(\sigma-1)} \cdot \frac{L_Y}{L},$$

where $\bar{A}$ captures weighted average productivity, N captures the variety/mass of firms, and L_Y/L is the share of labor producing current output rather than being used for entry, overhead, or research.

4.6 Endogenous life-cycle productivity

The paper finally considers models in which plants can invest in productivity growth. The cost of improving productivity is calibrated to match the U.S. life-cycle profile. The authors then introduce India-like or Mexico-like distortions in average revenue products.

The mechanism is:

- productive plants would like to grow and innovate;
- if productive plants face high wedges, innovation has lower private returns;
- lower post-entry innovation flattens the plant life cycle;
- aggregate TFP falls because the economy loses large, productive incumbents.

5 Identification and interpretation

5.1 What the paper identifies

The paper identifies robust differences in plant life-cycle dynamics across the United States, India, and Mexico. It then uses model-based accounting to quantify how important those differences could be for aggregate productivity.

The evidence is strongest as a set of empirical facts plus quantitative interpretation. The paper does not claim to isolate one unique causal barrier that prevents plant growth.

5.2 Why including small and informal plants matters

Plant-age analysis requires data that cover the full age and size distribution. If a data set only includes large plants, young small plants are mechanically underrepresented, making life-cycle growth hard to measure. This is why the paper emphasizes combining ASI with NSS in India and using census-type data for Mexico and the United States.

5.3 Age measurement concerns

Age is measured differently across countries. In the United States, age is inferred from first appearance in the Census. In India and Mexico, age is self-reported as the year production began. The paper discusses concerns such as ownership changes, plants becoming formal, and establishments without fixed locations. The exit-rate and robustness patterns reduce, but do not eliminate, these concerns.

5.4 Selection versus growth

Average employment can rise with age because surviving plants grow or because small plants exit. The paper compares current operating plants and survivors to show that selection alone cannot explain the U.S.-India contrast. In the United States, survivors grow substantially; in India, even survivors grow little or can shrink in some periods.

5.5 Revenue productivity as a distortion proxy

Revenue productivity is not a direct observation of a legal tax or a specific regulation. It is an accounting object that can reflect many forces: markups, taxes, financial constraints, management frictions, transportation costs, labor regulations, overhead costs, or adjustment costs.

The important pattern is that revenue productivity rises steeply with plant productivity in India and Mexico. This suggests that more productive plants are especially constrained or taxed at the margin, which can reduce their incentive to invest and expand.

5.6 Model-based interpretation

The GE simulations should be read as quantitative discipline rather than as structural proof. The models show that the observed life-cycle differences are large enough to matter for aggregate TFP, and that distortions against high-productivity plants can generate slow life-cycle growth. The exact magnitudes depend on model assumptions about entry, innovation costs, wedges, and product-market structure.

6 Tables and figures

6.1 Data coverage and first life-cycle facts

Read Table I and Figures I–III:

- Table I shows the core data sources and sample sizes. India has many more establishments and workers in the informal NSS than in the formal ASI, which is why informal coverage is central.
- Figure I shows the central cross-sectional fact: U.S. plants grow dramatically with age; Indian plants barely grow; Mexican plants grow initially but flatten.
- Figure II shows that exit rates in India and Mexico are not systematically higher than in the United States, so slow old-plant growth is not just a survival-rate story.
- Figure III shows that old plants account for almost 30 percent of U.S. manufacturing employment but less than 5 percent in India and Mexico.

Economic message: the United States has many large, old incumbents; India and Mexico are dominated much more by young and small plants.

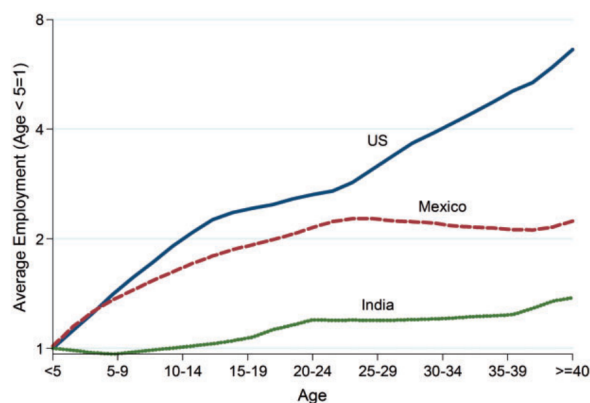


FIGURE I

Plant Employment by Age in the Cross-Section

Data from 2010–2011 ASI–NSS (India), 2003 Economic Census (Mexico), and the 2002 Manufacturing Census (United States). Employment in the youngest group (age < 5 years) is normalized to 1 in each country. The figure gives employment per operating plant versus plant age in the cross-section. In Mexico, employment includes paid and unpaid workers at fixed-location establishments. For the United States, employment covers all manufacturing establishments with at least one employee.

Figure 1: Figure I: Plant employment by age in the cross-section.

TABLE I
DATA

Indian Annual Survey of Industries and National Sample Survey

	# Establishments (raw data)		# Establishments (w/ sampling weights)		# Workers (w/ sampling weights)	
	ASI	NSS	ASI	NSS	ASI	NSS
1989–90	39	96	77	13,678	6,211	26,606
1994–95	45	157	93	12,153	6,953	20,357
2010–11	30	98	103	16,957	9,394	33,306

Mexican Economic Census

	# Establishments	# Workers
1998	344	4,226
2003	329	4,199
2008	437	4,661

United States Manufacturing Census

	# Establishments	# Workers
1992	371	16,949
1997	363	16,805
2002	351	14,664

Notes. Entries are in thousands. Data from the ASI–NSS (India), Economic Census (Mexico), and Manufacturing Census (United States). For India and Mexico, the number of workers include seasonal workers and contract workers.

Notes. Entries are in thousands. Data from the ASI–NSS (India), Economic Census (Mexico), and Manufacturing Census (United States). For India and Mexico, the number of workers include unpaid workers and contract workers.

Table 1: Table I: Data coverage for India, Mexico, and the United States.

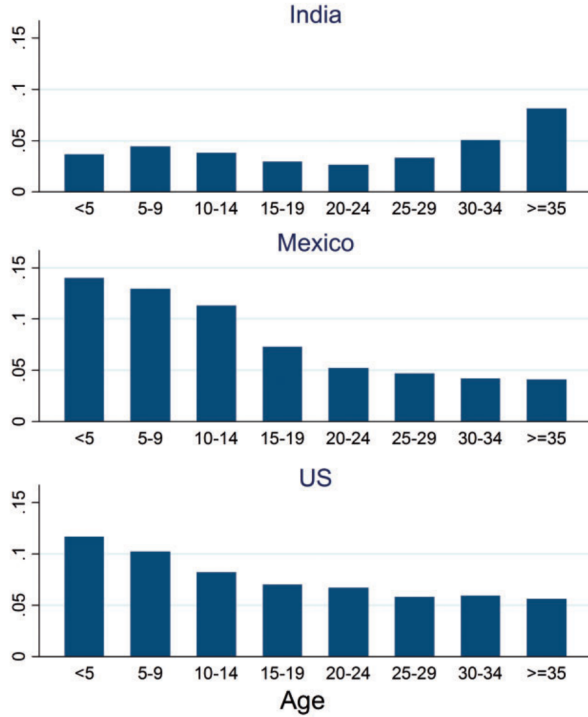


FIGURE II
Exit Rate by Age

Exit rates are calculated from 1992 to 1997 (U.S. Manufacturing Census), from 1994–1995 to 2010–2011 (Indian ASI-NSS), and 1998 to 2003 (Mexican Manufacturing Census). See text for details.

three countries (the spike in the last age group is due to pooling of

Figure 2: Figure II: Exit rate by age.

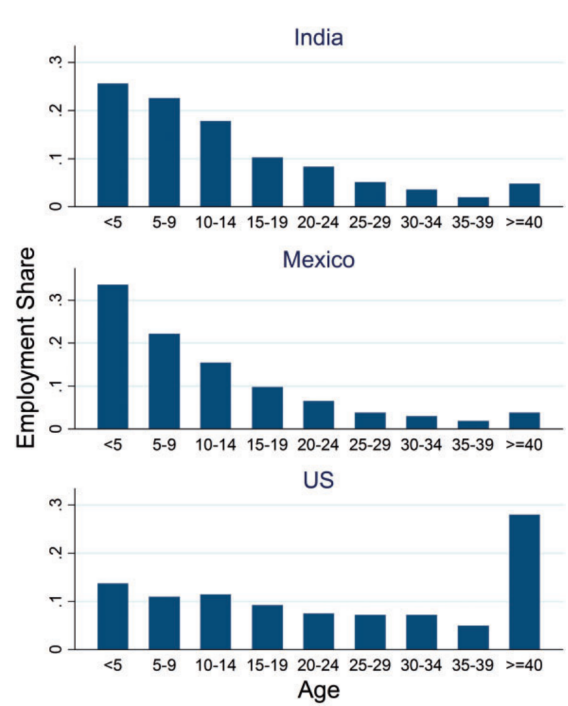


FIGURE III
Employment Share by Age in the Cross-Section

2010–2011 ASI-NSS (India), 2003 Economic Census (Mexico), and 2002 Manufacturing Census (United States). For India, employment includes paid, unpaid, and contract workers. In Mexico employment includes paid and unpaid workers at fixed-location establishments. For the United States, employment covers all manufacturing establishments with at least one employee.

Figure 3: Figure III: Employment share by age in the cross-section.

6.2 Cohort growth, entry cohorts, and survivor growth

Read Figures IV–IX:

- Figure IV follows cohorts over time and confirms the steep U.S. life cycle, modest Indian growth, and intermediate Mexican growth.
- Figure V shows that younger cohorts are larger in India and Mexico, which contributes to their age distribution.
- Figure VI constructs steady-state employment shares assuming constant entry and entrant size. Even under this benchmark, old plants matter much less in India and Mexico.
- Figure VII shows that India's life cycle improves after the early-1990s reforms: post-1994 cohorts grow, while pre-1994 cohorts shrink with age.
- Figure VIII separates growth of all operating plants from growth of survivors. Selection is not enough to explain the U.S. pattern.
- Figure IX adds other countries. The United States remains at the high-growth end and India at the low-growth end; Mexico is not uniquely slow among richer countries but remains far below the United States.

Economic message: the life-cycle gap is a persistent and economically meaningful fact, not an artifact of one cross-section.

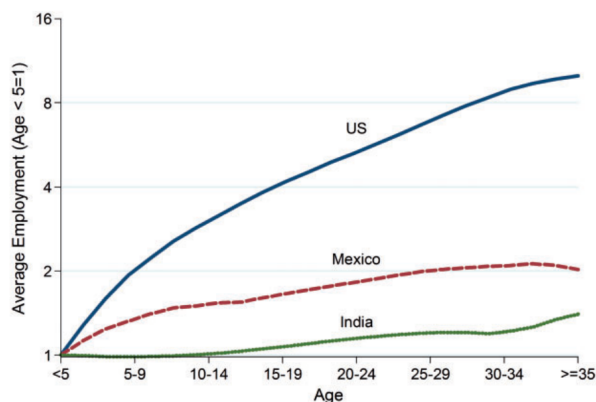


FIGURE IV

Average Employment over the Life Cycle

Employment growth imputed from the 1992 and 1997 U.S. Manufacturing Censuses, the 1998 and 2003 Mexican Economic Censuses, and the 1994–1995 and 2010–2011 Indian ASI-NSS. Employment of the youngest age group is normalized to 1 in each country. We compare average employment per surviving plant in a later year to average employment per operating plant in the same cohort in the earlier year. See text for details.

Figure 4: Figure IV: Average employment over the life cycle.

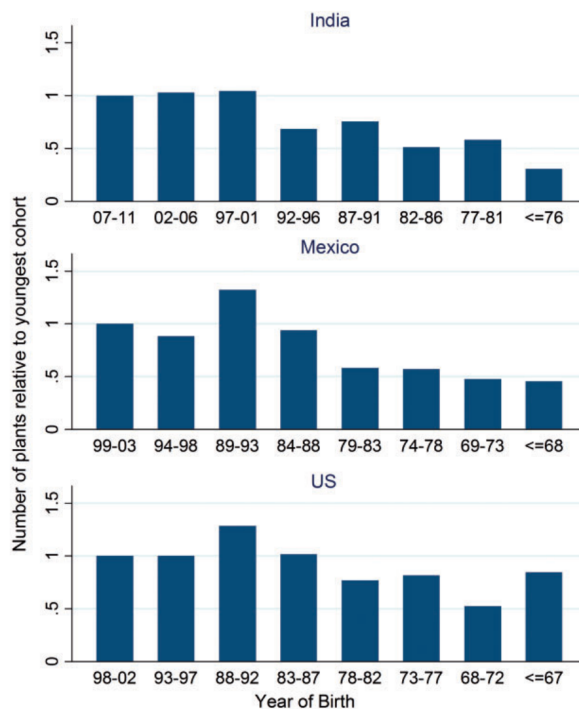


FIGURE V

Number of Plants by Birth Cohort

Cohort size imputed from cohort sizes in 2002 U.S. Manufacturing Census, 2008 Mexican Economic Census, and 2010–2011 Indian ASI-NSS assuming that exit rates by age (shown in Figure II) are the same for all cohorts.

2010–2011 in India). We then impute the size of each cohort at

Figure 5: Figure V: Number of plants by birth cohort.

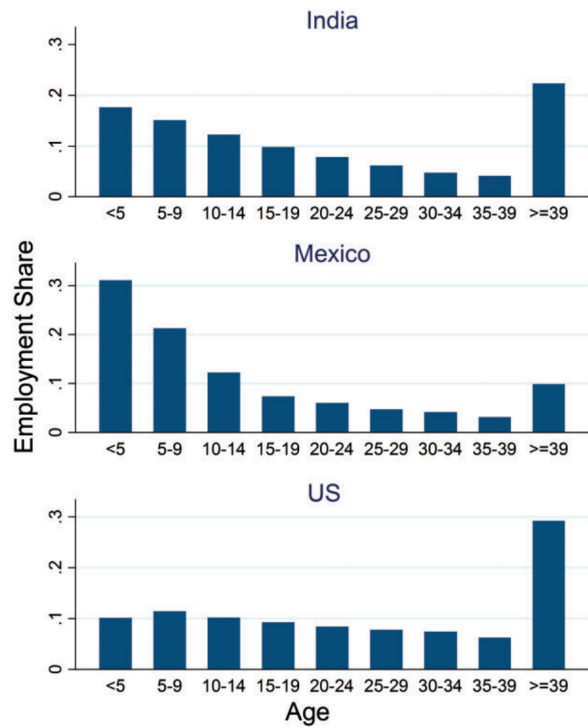


FIGURE VI

Employment Share by Age in Steady State

Steady-state employment share by age assumes constant entry rate and constant average size of entrants. The figure assumes growth in average employment per surviving plant as given by Figure IV and exit rate by age as given by Figure II.

Figure 6: Figure VI: Employment share by age in steady state.

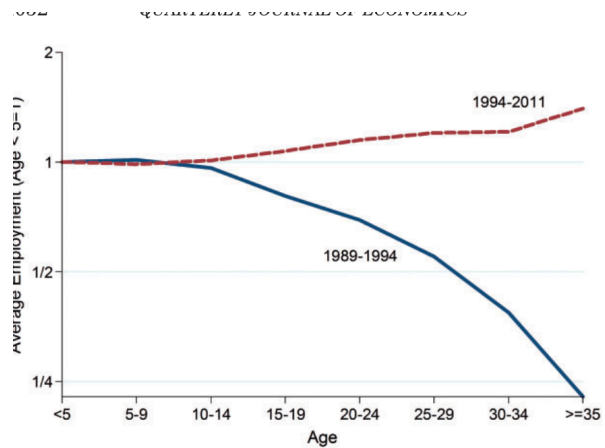


FIGURE VII

Employment Growth in India

Employment growth with age imputed from the ASI and NSS from 1989–990 to 1994–1995, and from 1994–1995 to 2010–2011.

In contrast, the evidence from Mexico and the United States

Figure 7: Figure VII: Employment growth in India.

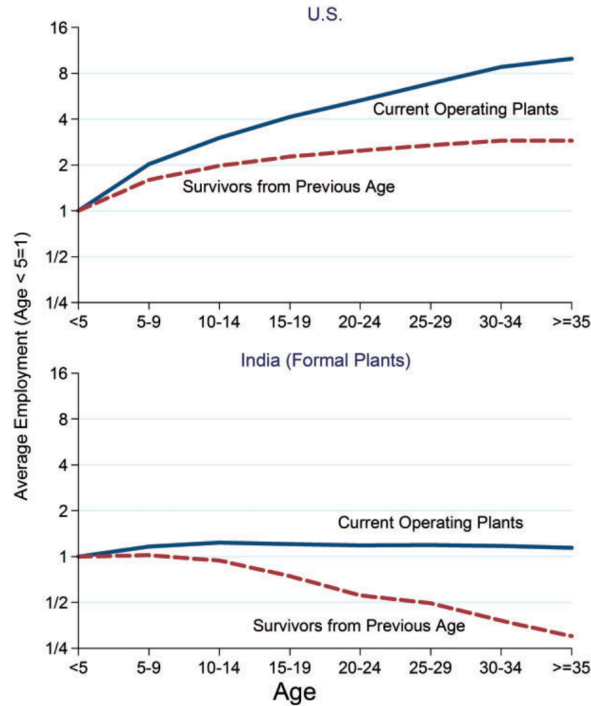


FIGURE VIII
Employment Growth over the Life Cycle

Employment growth with age is imputed from the 1992 and 1997 U.S. Manufacturing Censuses and the 1998–1999 and 2003–2004 Indian ASI. “Survivors from Previous Age” are based on comparing plants that operate in both of the years. “Current Operating Plants” is based on all operating plants, including those who do not survive to the latter year.

Figure 8: Figure VIII: Employment growth over the life cycle, survivors versus current plants.

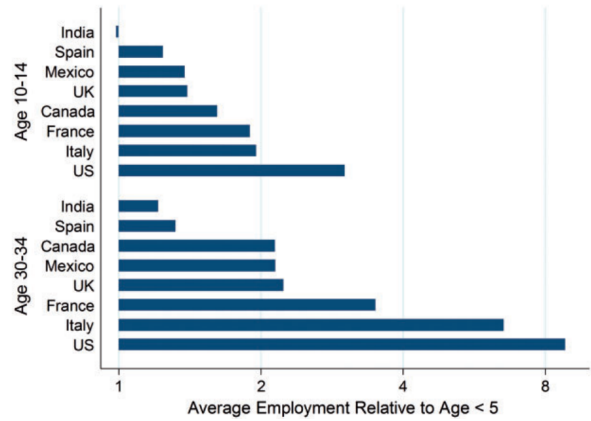


FIGURE IX

Employment Growth over the Life Cycle

Employment growth by age 10–14 and age 30–34 relative to age < 5. Indian data are from plants in the 2009–2010 ASI/NSS. Data for France, Italy, and Spain are for firms in the 2006–2007 Amadeus Database. U.K. data are for plants from 1997–2001 to 2002–2006 in the ARD. Canadian data are for plants from 1999–2001 to 2004–2006 in the Canadian ASM. See Appendix I for additional details.

Figure 9: Figure IX: Cross-country employment growth over the life cycle.

6.3 Productivity, revenue productivity, and distortions

Read Figures X–XIII:

- Figure X shows that productivity grows much more over the life cycle in the United States than in India or Mexico. Slow employment growth reflects slow productivity growth, not merely different input choices.
- Figure XI shows average revenue products over age. Indian old plants have slightly higher revenue productivity than young plants, while U.S. and Mexican old plants have slightly lower revenue productivity.
- Figure XII is crucial: in India and Mexico, revenue productivity rises steeply with plant productivity. More productive plants appear to face higher effective wedges.
- Figure XIII illustrates the model’s logic: if incumbents in one country accumulate productivity much faster after entry, average productivity at a point in time will be higher.

Economic message: the data suggest that India’s and Mexico’s more productive plants may be disproportionately constrained, which weakens post-entry productivity accumulation.

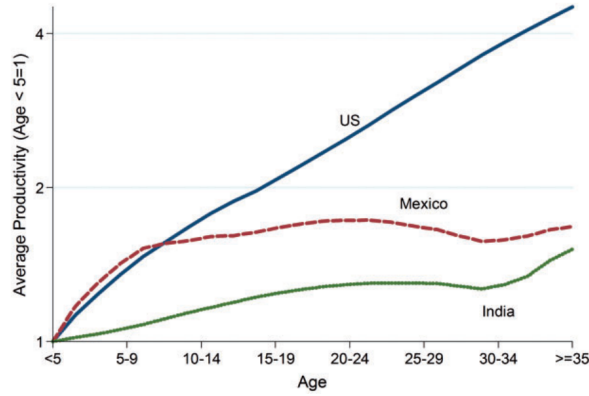


FIGURE X
Productivity over the Life Cycle

Productivity growth imputed from the 1992 and 1997 U.S. Manufacturing Censuses, the 1998 and 2003 Mexican Economic Censuses, and the 1994–1995 to 2010–2011 ASI and NSS in India. Productivity of the youngest age group is normalized to 1 in each country. We compare average productivity per surviving plant relative to aggregate TFP in a later year to average productivity per operating plant relative to aggregate TFP in the same cohort in the earlier year.

Figure 10: Figure X: Productivity over the life cycle.

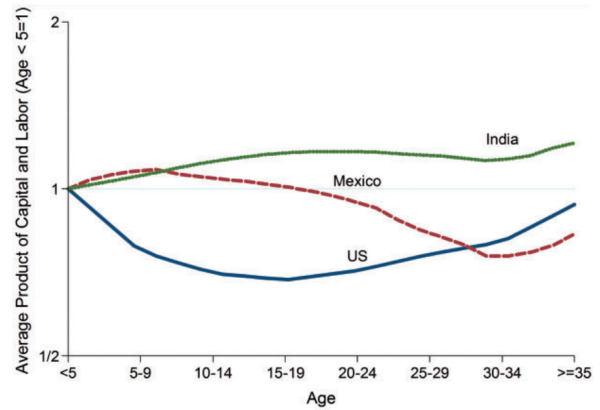


FIGURE XI
Revenue Productivity over the Life Cycle

Growth of the average revenue product of capital and labor in the 1992 and 1997 U.S. Manufacturing Censuses, the 1998 and 2003 Mexican Economic Censuses, and the 1994–1995 and 2010–2011 ASI-NSS in India. For the youngest age, Revenue Productivity is normalized to 1.

Figure 11: Figure XI: Revenue productivity over the life cycle.

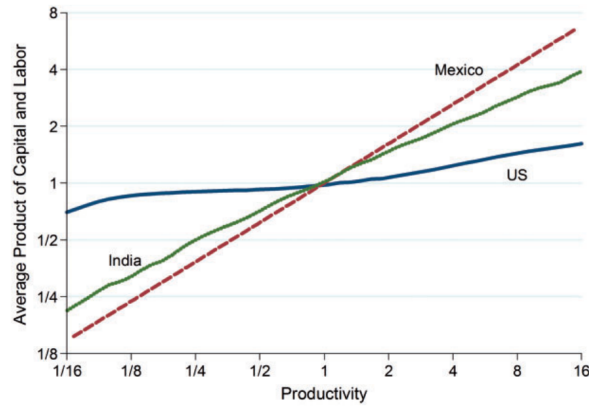


FIGURE XII
Revenue Productivity versus Productivity in the Cross-Section

The average revenue product of capital and labor (τ) and productivity (A) are relative to weighted averages of industry τ and A in each country. Sources: 2010–2011 ASI-NSS (India), 2003 Economic Census (Mexico), and 1992 Manufacturing Census (United States).

Figure 12: Figure XII: Revenue productivity versus productivity in the cross-section.

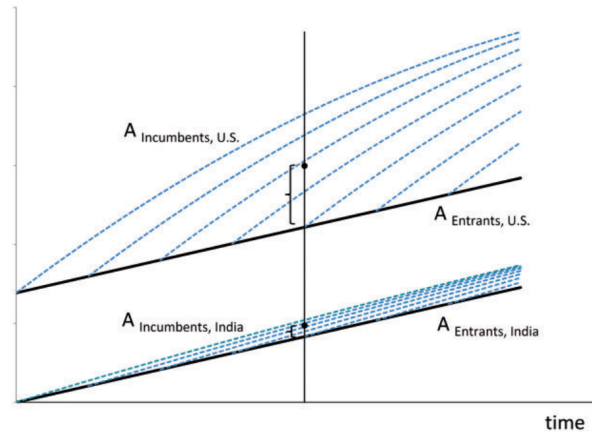


FIGURE XIII
Hypothetical Productivity across Cohorts

Figure 13: Figure XIII: Hypothetical productivity across cohorts.

6.4 Exogenous life-cycle-productivity models

Read Tables II–V and Figure XIV:

- Table II lists the exogenous life-cycle-productivity model variants: fixed versus free entry, fixed versus endogenous entrant quality, and different sources of average-product variation.
- Table III gives common parameter targets, including $\sigma = 3$, annual real interest rate of 5 percent,

and a 100-year maximum firm lifespan.

- Tables IV and V report aggregate TFP losses from replacing the U.S. life-cycle productivity profile with Indian or Mexican profiles. Baseline losses are about 25 percent for India and 18 percent for Mexico.
- Free entry raises the number of firms and partially offsets the TFP loss, but the weighted average productivity loss remains large.
- Figure XIV shows that India and Mexico have many more very small establishments than the United States.

Economic message: even simple models imply that slow life-cycle productivity growth can have first-order aggregate productivity consequences.

TABLE II
MODELS WITH EXOGENOUS LIFE CYCLE PRODUCTIVITY

Rows in Tables IV and V	Entry	Entrant quality	τ variation
Row 1	Fixed	Fixed	None
Row 2	Free	Fixed	None
Row 3	Free	Endogenous	None
Row 4	Free	Fixed	Overhead costs
Row 5	Free	Fixed	Adjustment costs
Row 6	Free	Fixed	Revenue taxes

Notes: A refers to firm productivity, and τ to the firm's average product of capital and labor. In all cases there is dispersion of A within and across ages, and exit is exogenous and varies by age and productivity.

Table 2: Table II: Models with exogenous life-cycle productivity.

TABLE III
PARAMETER VALUES FOR EXOGENOUS LIFE CYCLE PRODUCTIVITY

Parameter	Definition	Value or target
σ	Elasticity of substitution between varieties	3
γ	Coefficient of relative risk aversion	2
ρ	Discount rate	Annual real interest rate of 5%
N	Maximum life span of a firm	100 years (20 periods; 1 period = 5 years)
g_e	Growth rate of mean entrant A	2.1% per year for all models (U.S. average TFP growth)
A_a	Productivity by age	Set to match productivity by age data in each country
σ_e	Std. dev. of entrant log productivity	1.01 to match productivity dispersion of age 0–5 U.S. plants
δ_a	Exit by age, conditional on productivity	U.S. exit rates by 5-year age group
$\delta_{a,i}$	Exit by productivity, conditional on age	U.S. semi-elasticity of exit w.r.t. plant productivity
f_e	Entry costs (in terms of labor)	Average workers per plant in the U.S.
$\tau_{a,i}$	Average products by productivity level	Set to match U.S. elasticity of average products w.r.t. productivity

Notes: Average products vary with productivity only in the presence of overhead costs, adjustment costs, or revenue taxes (rows 4–6 in Tables II, IV, and V). See Section V and the Online Appendix for more detail.

Table 3: Table III: Parameter values for exogenous life-cycle productivity.

TABLE IV
PERCENT CHANGE FROM U.S. TO INDIAN LIFE CYCLE IN MODELS WITH EXOGENOUS LIFE CYCLE PRODUCTIVITY

Cases	Aggregate TFP	Weighted average A	Entry	Workers/workforce
Baseline	−25.1	−25.1	0	0
Free entry	−23.3	−25.1	+11.3	−3.6
Endogenous entrant quality	−28.9	−46.4	+100.7	0
Overhead costs	−16.5	−24.8	+26.1	−2.6
Adjustment costs	−22.3	−24.6	+12.5	−3.5
Revenue taxes	−23.4	−25.1	+9.9	−3.2

Notes: Table entries are % changes when going from U.S. to Indian productivity (A) by age. Aggregate TFP is the product of three terms ($TFP = \frac{Y}{L} = \bar{A} N^{\frac{1}{1+\sigma}} \frac{L}{L}$), weighted average A , a variety term involving the mass of firms, and the fraction of the population producing current output (as opposed to supplying overhead labor or generating entry).

Table 4: Table IV: Percent change from U.S. to Indian life cycle in exogenous models.

TABLE V
PERCENT CHANGE FROM U.S. TO MEXICAN LIFE CYCLE IN MODELS WITH EXOGENOUS LIFE CYCLE PRODUCTIVITY

Cases	Aggregate TFP	Weighted average A	Entry	Workers/workforce
Baseline	−18.2	−18.2	0	0
Free entry	−16.7	−18.2	+7.8	−2.5
Endogenous entrant quality	−23.5	−42.4	+101.5	0
Overhead costs	−13.0	−17.9	+14.9	−2.1
Adjustment costs	−16.0	−18.0	+9.3	−2.7
Revenue taxes	−16.8	−18.2	+6.7	−2.2

Notes: Table entries are % changes when going from U.S. to Mexican productivity (A) by age. Aggregate TFP is the product of three terms ($TFP = \frac{Y}{L} = \bar{A} N^{\frac{1}{1+\sigma}} \frac{L}{L}$), weighted average A , a variety term involving the mass of firms, and the fraction of the population producing current output (as opposed to supplying overhead labor or generating entry).

Table 5: Table V: Percent change from U.S. to Mexican life cycle in exogenous models.

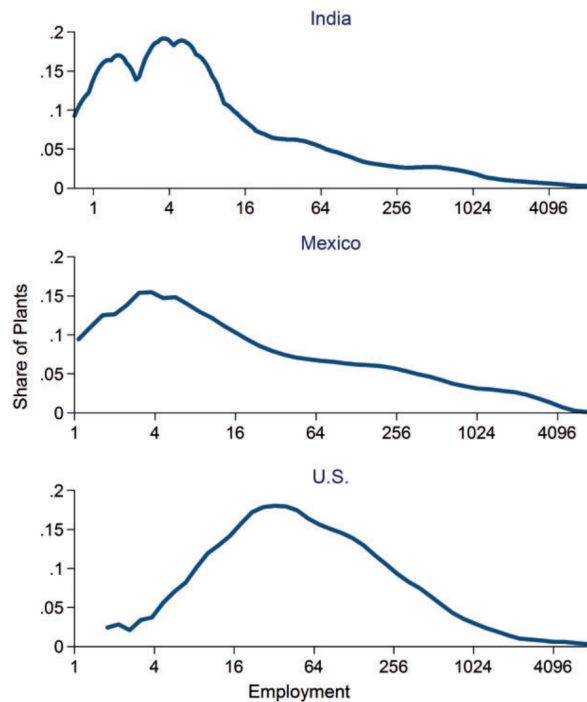


FIGURE XIV

Distribution of Establishments by Employment

Sources: 2010–2011 ASI-NSS (India), 1998 Economic Census (Mexico), and 1997 Manufacturing Census (United States). Plants are weighted by the per-plant value-added share of each four-digit industry.

entry costs in units of labor in light of Rollard, Klenow, and Li

Figure 14: Figure XIV: Distribution of establishments by employment.

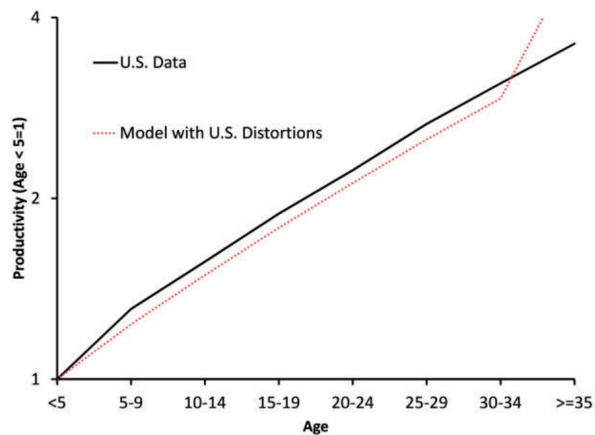


FIGURE XV

Productivity by Age in the United States, Data and Models

Figure 15: Figure XV: Productivity by age in the United States, data and model.

6.5 Endogenous productivity and distortion experiments

Read Table VI, Figures XVI–XVII, and Table VII:

- Table VI gives the endogenous-productivity parameters. Plants invest in productivity subject to an R&D cost function, and average products vary with productivity.
- Figure XV shows that the model can fit the U.S. productivity-age profile.
- Figures XVI and XVII compare data and model profiles when Indian or Mexican distortions are imposed. The model captures the idea that stronger distortions flatten life-cycle productivity growth.
- Table VII reports large TFP effects when moving from U.S. to Indian or Mexican average-product schedules: aggregate TFP falls by 36.5 percent for India and 53.0 percent for Mexico in the reported endogenous-productivity exercise.

Economic message: when high-productivity plants face steeper wedges, they invest less in productivity growth. This is a dynamic amplification of misallocation.

TABLE VI
PARAMETER VALUES FOR ENDOGENOUS LIFE CYCLE PRODUCTIVITY

Parameter	Definition	Value or target
h	Level parameter in the R&D cost function	Set with b to match U.S. productivity by age group
b	Convexity parameter in the R&D cost function	Set with h to match U.S. productivity by age group
$\tau_{a,i}$	Average products by productivity level	Set to match elasticity of average products w.r.t. productivity in each country

Notes. The following parameters are the same as in the simulations with exogenous productivity by age (see Table III): $\sigma=3$ (the elasticity of substitution between varieties); $\gamma=2$ (the coefficient of relative risk aversion); ρ (the discount rate) to yield an annual real interest rate of 5%; $N=100$ (maximum life span of a firm in years); $g_e=0.021$ (growth of mean entrant productivity); $A_{0,i}$ (entrant productivity dispersion) to match productivity dispersion of young U.S. plants; δ_i (exit rate by age conditional on productivity) to match U.S. exit by age data; $\delta_{a,i}$ (exit rate by productivity for a given age) to match U.S. semi-elasticity of exit w.r.t. productivity; and f_e (entry costs in terms of labor) to match average employment per plant in the United States. Average products vary with productivity due to revenue taxes. Entry is free, productivity growth is stochastic and endogenous, and there are no overhead costs or adjustment costs. See Section V and Appendix I for more detail.

Table 6: Table VI: Parameter values for endogenous life-cycle productivity.

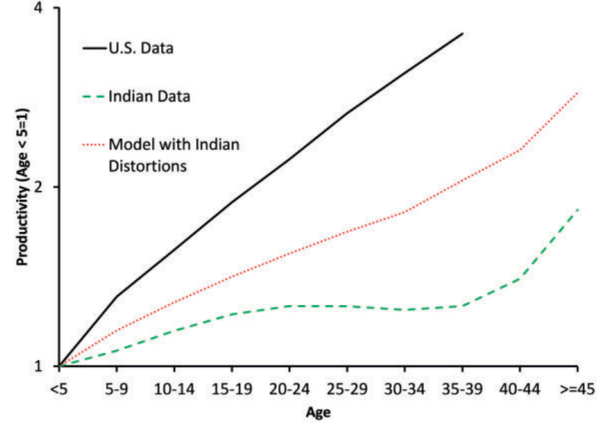


FIGURE XVI
Productivity by Age in India, Data and Models

Figure 16: Figure XVI: Productivity by age in India, data and model.

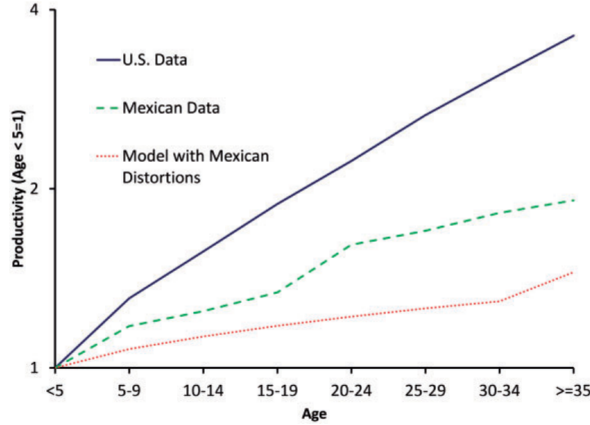


FIGURE XVII
Productivity by Age in Mexico, Data and Models

Figure 17: Figure XVII: Productivity by age in Mexico, data and model.

TABLE VII PERCENT CHANGE FROM THE U.S. TO INDIAN AND MEXICAN AVERAGE PRODUCTS IN MODELS WITH ENDOGENOUS LIFE CYCLE PRODUCTIVITY				
Cases	Aggregate TFP	Weighted average A	Entry	Workers/workforce
India	-36.5	-39.7	+14.3	-1.5
Mexico	-53.0	-55.7	+16.4	-1.3

Notes. Table entries are % changes when going from U.S. to Indian average products (τ) by age. Aggregate TFP is the product of three terms ($TFP = \tau = A N^{\frac{1}{1-\sigma}} \frac{1}{L^{\frac{\sigma}{1-\sigma}}}$): weighted average A , a variety term involving the mass of firms, and the fraction of the population producing current output (as opposed to supplying overhead labor, generating entry, or doing research).

Table 7: Table VII: TFP effects in endogenous life-cycle-productivity models.

6.6 Informality and appendix data checks

Read Table VIII and Figure A.I:

- Table VIII shows that informality is central in India and still important in Mexico. In India, informal establishments account for the vast majority of establishments and a majority of workers.
- Figure A.I compares the Indian ASI-NSS employment-size distribution to the 2005 Economic Census. The two sources have similar qualitative size distributions, supporting the representa-

tiveness of the combined data.

Economic message: small and informal establishments are not peripheral. They are central to the manufacturing structure in India and Mexico, so any analysis of plant life cycles must include them.

TABLE VIII
INFORMAL WORKERS AND ESTABLISHMENTS IN INDIA AND MEXICO

	Unpaid workers		Informal establishments	
	% Workers	% Establishments	% Workers	% Establishments
India				
1989-90	73.4	94.2	80.5	99.4
2010-11	54.2	90.8	78.5	99.4
Mexico				
1998	10.2	55.0	14.8	75.6
2008	29.7	60.0	30.4	87.1

Notes. “% Workers” is the percent of unpaid workers or workers in informal establishments as a share of total workers. “% Establishments” is percent of establishments with only unpaid workers or that are informal as share of total number of establishments. Informal establishments are defined as establishments not formally registered (in India) or not registered with the Social Security Agency (in Mexico).

Table 8: Table VIII: Informal workers and establishments in India and Mexico.

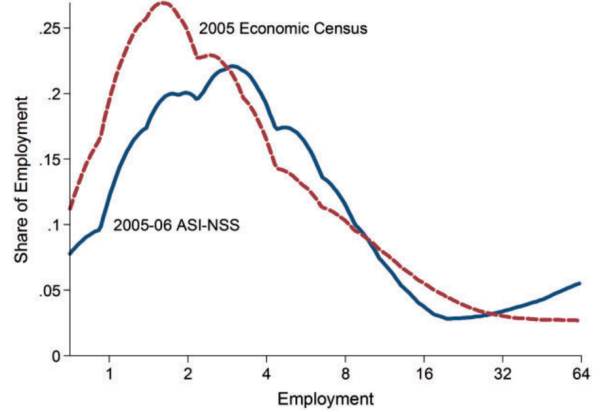


FIGURE A.I
Distribution of Employment by Establishment Size (2005 Census versus 2005–2006 ASI-NSS)

Figure 18: Figure A.I: Indian employment-size distribution, ASI-NSS versus Economic Census.

7 Short summary

- The paper studies why plants grow much less with age in India and Mexico than in the United States.
- The main empirical fact is that old U.S. manufacturing plants are dramatically larger than young U.S. plants, while old Indian and Mexican plants are only modestly larger.
- This fact is not mainly driven by higher exit rates in India or Mexico.
- Old plants account for a much smaller share of employment in India and Mexico than in the United States.
- Cohort-based evidence confirms the cross-sectional fact, especially for the United States and Mexico.
- India’s post-1994 life-cycle growth is stronger than its pre-1994 pattern, but still far below the United States.
- Productivity grows much more over the plant life cycle in the United States than in India or Mexico.
- More productive plants in India and Mexico have much higher average revenue products, suggesting that they face higher wedges or barriers to expansion.
- Simple GE models imply that moving from the U.S. life cycle to the Indian or Mexican life cycle can lower aggregate manufacturing productivity by about 25 percent.

- Endogenous-productivity models show how distortions against productive plants can reduce the incentive to invest in post-entry productivity growth.
- The paper identifies barriers to plant growth as a potentially important source of cross-country TFP differences.

8 Conclusion

8.1 Core conclusion

Hsieh and Klenow show that the plant life cycle differs sharply across countries. U.S. manufacturing plants grow enormously as they age, while plants in India and Mexico grow much less. This difference is large enough to matter for aggregate productivity: slow life-cycle growth can plausibly lower manufacturing TFP by around a quarter relative to the United States.

8.2 Economic intuition

Plants do not become productive automatically. They invest in organizational capital, managerial systems, product quality, technology adoption, customer relationships, and access to markets. If large or productive plants face high marginal barriers, the payoff to those investments falls. Firms stay small, entrants face less competition, and aggregate productivity remains low.

8.3 Contribution revisited

The paper extends the misallocation literature from a static question to a dynamic one. Misallocation does not only put too few inputs into productive plants today. It can also prevent plants from becoming highly productive tomorrow. This dynamic effect helps explain why average plant productivity is lower in poorer economies.

8.4 Implications

The findings imply that development policy should focus not only on creating new firms but also on allowing successful firms to grow. Reducing barriers that bind more tightly on large and productive establishments may have high returns because it raises both current input allocation and future productivity investment.

8.5 Limitations

The paper does not identify a single causal barrier behind slow life-cycle growth. Revenue-productivity wedges are reduced-form objects that can represent many forces. Age measurement and informal-sector coverage also differ across countries. Finally, the quantitative models are intentionally simple, so the exact TFP magnitudes should be interpreted as disciplined calculations rather than definitive structural estimates.

8.6 Takeaway

The enduring takeaway is that plant dynamics are central to productivity differences. Economies with many small firms are not necessarily entrepreneurial success stories if productive plants cannot scale. Development requires not only entry, but also the growth and upgrading of the best incumbents.